

Physical Optics

According to the Newton's corpuscular theory, a luminous body continuously emits invisible, tiny, light and elastic particles called corpuscles. These corpuscles are emitted in all directions with tremendous velocity. Since the corpuscles are emitted with high velocity, they, in the absence of other forces, travel in straight lines according to Newton's law of motion, which explains rectilinear propagation of light. Their mechanical impact on the retina produces the sensation of vision. The different colors were attributed to the different sizes of the corpuscles. In reflection, the particles were supposed to act like small elastic spheres reflected from the surface like marble balls from a concrete floor. In refraction, however, it was necessary to assume that the component of the velocity perpendicular to the surface suddenly increased if the second medium was denser than the first one. But, this corpuscular theory of light could not successfully explain the phenomenon of interference, diffraction and polarization of light (the other three important characters of light).

Huygens first put the wave theory of light forward in 1678. Huygens wave model could explain the above characteristic phenomenon of light (interference, diffraction and polarization of light). On the basis of wave theory. Huygens also explained the phenomena of reflection, refraction and total internal refraction. Huygens' theory predicted that the velocity of light in a medium (like water) should be less than the velocity of light in free space which is just the converse of the prediction, made from Newton's corpuscular theory.

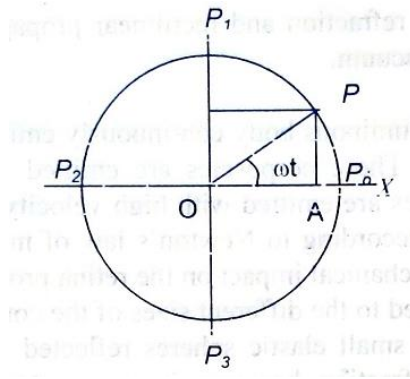
Huygens' theory assumes that light is a wave, but it says nothing about the nature of wave. Huygens did not know whether light a transverse wave or a longitudinal one was; he did not know the wavelengths of visible light; he had little knowledge of the speed of light. Nevertheless, his theory was a useful guide to experiment for many years and remains useful today for pedagogic and certain other practical purposes. However, for the better understanding of Huygens principle, the concept of wave motion is of great importance.

Wave Motion

Wave motion is a form of disturbance, which travels through a medium due to the repeated periodic motion of the particles of the medium about their mean position, the motion being handed over from one particle to the next".

In a wave motion, the particles of the medium execute simple harmonic vibrations (oscillations) about their mean position along or perpendicular to the direction of propagation of the wave according to, as it is a longitudinal or a transverse wave. The disturbance is handed on from particle to particle after a definite time and there is a gradual fall of phase in the direction of motion.

In order to understand the wave motion, let us consider a reference particle at point P rotating on the circumference of a circle of radius a with an angular velocity (J) as shown in figure. We choose the center of the circle as our origin and assume that at $t = 0$ the particle at P lies on the X-axis (i.e., at point P_0). At an arbitrary time t the particle will be at the position P where $\angle POX = \omega t$.



Let A be the foot of the perpendicular from the particle at point, P on the X-axis. Clearly the distance,

$$OA = a \cos \omega t$$

$$\Leftrightarrow x = a \cos \omega t$$

Here, the quantity a is called the amplitude of the wave motion and the period of motion, T will be the time required to complete the one revolution. If ω be the angular frequency then the time period will be –

$$T = \frac{2\pi}{\omega}$$

The inverse time period is called the frequency.

$$f = \frac{1}{T} = \frac{\omega}{2\pi}$$

$$\omega = 2\pi f$$

We also can write the wave equation as like –

$$x = a \cos(\omega t + \delta)$$

Here, the quantity $(\omega t + \delta)$ is known as the phase of the wave motion and δ is the initial phase or phase difference.

Differential Equation of the Wave Motion:

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

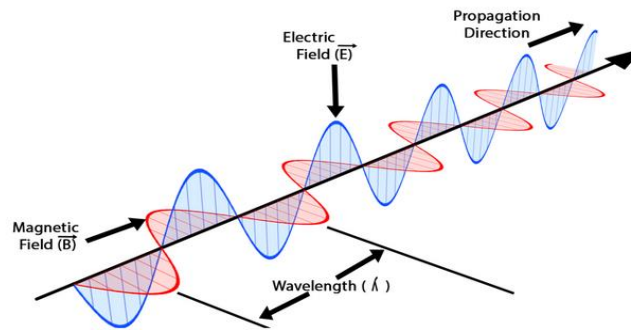
Huygens supposed that the light waves were longitudinal but Fresnel and Young later explained that these were transverse in nature.

In 1860, **Clark Maxwell** introduced an idea of electromagnetic theory of light. According to this theory, the light waves are composed of varying, coupled electric and magnetic fields perpendicular to one another and the direction of propagation of waves. The velocity of the propagation of electromagnetic wave is given by –

$$v = \frac{1}{\sqrt{K\mu}} \quad \text{Where, } K \text{ is the dielectric constant and } \mu \text{ is}$$

the permeability of the medium. In empty space, the velocity of EM wave is $3 \times 10^8 \text{ ms}^{-1}$.

Electromagnetic Wave



Electromagnetic waves carry energy as they travel through empty space. There is an energy density associated with both the electric field E and the magnetic field B . The rate of energy transport per unit area is described by the vector –

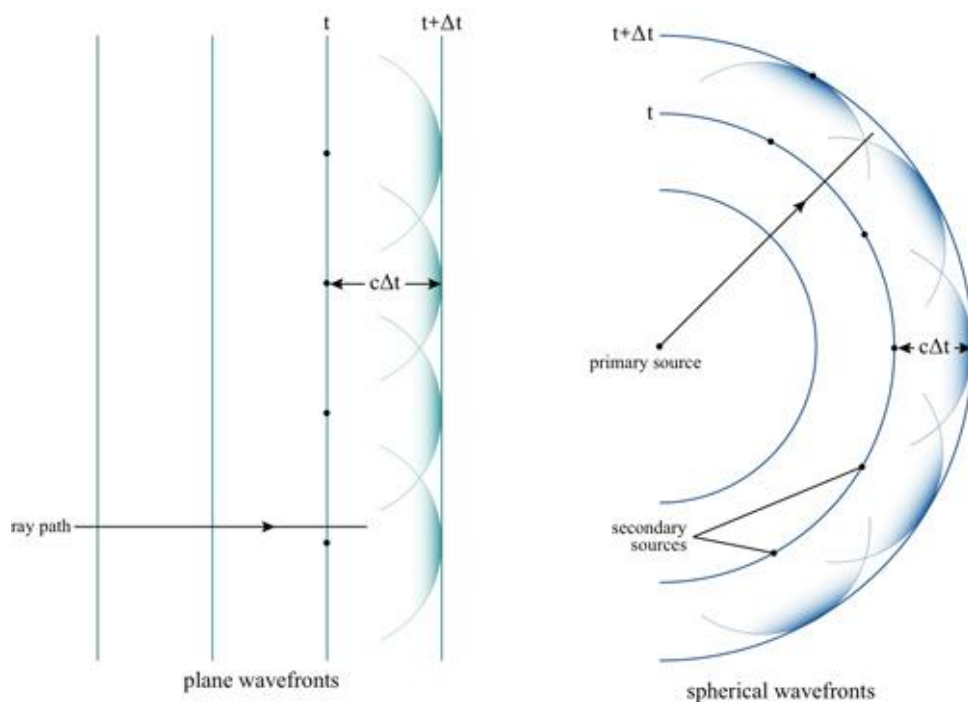
$$\vec{S} = \frac{1}{\mu} \vec{E} \times \vec{B}$$

which is called the **Poynting vector**. This is a vector quantity and the direction of it is the propagation direction of the electromagnetic wave.

Wave Front and Huygens Principle:

Wave front, imaginary surface representing corresponding points of a wave that vibrate in unison. When identical waves having a common origin travel through a homogeneous medium, the corresponding crests and troughs at any instant are in phase; i.e., they have completed identical fractions of their cyclic motion, and any surface drawn through all the points of the same phase will constitute a wave front.

We can define in another way that, the **wave front** is the locus of the point, which are in the same point. So, the phase differences between two waves in a wave front is zero.

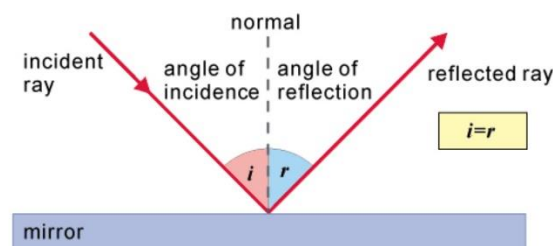


Huygens had a very important insight into the nature of wave propagation which is now a days called Huygens' principle. When applied to the propagation of light waves, this principle states that:

“Every point on a wave-front may be considered a source of secondary spherical wavelets which spread out in the forward direction at the speed of light. The new wave-front is the tangential surface to all of these secondary wavelets.”

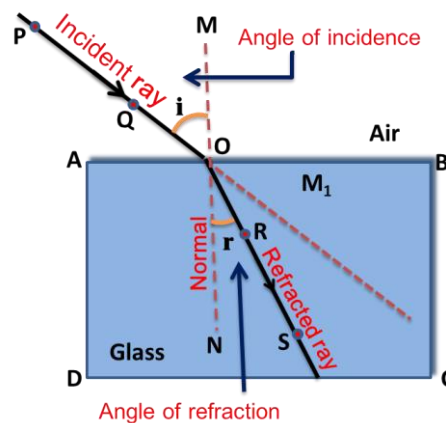
Laws of Reflection:

1. The first law of reflection states that the incident ray, the reflected ray, and the normal to the surface of the mirror, all lie in the same plane.
2. The second law of reflection states that the angle of reflection is equal to the angle of incidence. Both angles are measured with respect to the normal to the mirror.



Laws of Refraction:

1. The incident ray, the refracted ray and the normal to the interface of two transparent media at the point of incidence, all lie in the same plane.



2. The ratio of sine of angle of incidence to the sine of angle of refraction is a constant, for the light of a given color and for the given pair of media. This law is also known as Snell's law of refraction.

$$\mu = \frac{\sin i}{\sin r}$$

This constant, μ is also called the refractive index of the second medium with respect to the first.

It is possible to derive the laws of reflection and refraction of light by using Huygens Principle.